

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	N. Tharun Sai	Reg. No.:	192472382
Section / Batch:	CSE-AI	Date:	

Code of Conduct

I, N. Tharun Sai / 192472382 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Tharun Sai

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

```
import re
```

```
# Student Details
```

```
email = "tharunsai123@gmail.com"
```

```
mobile = "+91-9876543210"
```

```
password = "P@ssw0rd123"
```

```
dob = "15/08/2004"
```

```
register_number = "23A1ML1056"
```

```
# Regex Patterns
```

```
email_pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
```

```
mobile_pattern = r'^\+91-[6-9]\d{9}$'
```

```
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$'
```

```
dob_pattern = r'^([0-1-9]|[12][0-9]|3[01])/([0-1-9]|1[0-2])/([0-4])$'
```

```
register_pattern = r'^\d{2}[A-Z]{4}\d{4}$'
```

```
# Validation
```

```
print("Email:", "Valid" if re.fullmatch(email_pattern, email) else "Invalid")
```

```
print("Mobile:", "Valid" if re.fullmatch(mobile_pattern, mobile) else "Invalid")
```

```
print("Password:", "Valid" if re.fullmatch(password_pattern, password) else "Invalid")
```

```
print("Date of Birth:", "Valid" if re.fullmatch(dob_pattern, dob) else "Invalid")
```

```
print("Register Number:", "Valid" if re.fullmatch(register_pattern, register_number) else "Invalid")
```

```
*** Email: Valid
```

```
Mobile: Valid
```

```
Password: Valid
```

```
Date of Birth: Valid
```

```
Register Number: Valid
```

```
3. Extract words
```

```
# Given passage
```

```
pass = "Intelligence (AI) is transforming industries across the world.
```

```
It is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences.
```

```
With constant threat levels, it is research because it improves efficiency and enables intelligent decision-making.
```

```
As AI continues to evolve, professionals with AI skills are in high demand!"
```

```
# 1. Split the passage into sentences using re.split()
```

```
sentences = re.split(r'\.!', pass)
```

```
# Remove empty strings
```

```
sentences = [s.strip() for s in sentences if s.strip()]
```

```
# 2. Count total number of sentences
```

```
print("Total number of sentences:", len(sentences))
```

```
# 3. Display the extracted sentences
```

```
print("Extracted sentences:")
```

```
for i, sentence in enumerate(sentences, start=1):
```

```
print(f"{i}. {sentence}")
```

```
# 4. Split the passage into words using re.findall() to extract individual characters
```

```
words = re.findall(r'\w+', pass.lower())
```

```
# 5. Count total number of words
```

```
print("Total number of words:", len(words))
```

```
# 6. Display the extracted words
```

```
print("Extracted words:")
```

```
print(words)
```

```
# 7. Sentences :
```

```
passage:
```

```
Intelligence (AI) is transforming industries across the world.
```

```
It is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences.
```

```
With constant threat levels, it is research because it improves efficiency and enables intelligent decision-making.
```

```
As AI continues to evolve, professionals with AI skills are in high demand!"
```

```
Words:
```

```
Intelligence, (AI), is, transforming, industries, across, the, world, it, is, used, in, healthcare, to, assist, doctors, in, diagnosis, in, banking, to, detect, fraud, and, in, education, to, provide, personalized, learning, experiences, it, is, research, because, it, improves, efficiency, and, enables, intelligent, decision-making, As, AI, continues, to, evolve, professionals, with, AI, skills, are, in, high, demand,
```

```
Words:
```

```
Intelligence, (AI), is, transforming, industries, across, the, world, it, is, used, in, healthcare, to, assist, doctors, in, diagnosis, in, banking, to, detect, fraud, and, in, education, to, provide, personalized, learning, experiences, it, is, research, because, it, improves, efficiency, and, enables, intelligent, decision-making,
```

```
Words:
```

```
Intelligence, (AI), is, transforming, industries, across, the, world, it, is, used, in, healthcare, to, assist, doctors, in, diagnosis, in, banking, to, detect, fraud, and, in, education, to, provide, personalized, learning, experiences, it, is, research, because, it, improves, efficiency, and, enables, intelligent, decision-making,
```

```
Words:
```

```
Intelligence, (AI), is, transforming, industries, across, the, world, it, is, used, in, healthcare, to, assist, doctors, in, diagnosis, in, banking, to, detect, fraud, and, in, education, to, provide, personalized, learning, experiences, it, is, research, because, it, improves, efficiency, and, enables, intelligent, decision-making,
```

```
Words:
```

```
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```

```
Words:
```

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```

```
Words:
```

```
Intelligence, (AI), is, transforming, industries, across, the, world, it, is, used, in, healthcare, to, assist, doctors, in, diagnosis, in, banking, to, detect, fraud, and, in, education, to, provide, personalized, learning, experiences, it, is, research, because, it, improves, efficiency, and, enables, intelligent, decision-making,
```



```
import re

# Read input
email = input("Enter Email ID: ")
password = input("Enter Password: ")

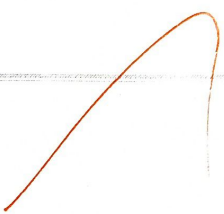
# Email Regex
email_pattern = r'^[A-Za-z0-9][A-Za-z0-9._%+-]*@[A-Za-z0-9-]+\.(com|org|edu|in)$'

# Password Regex
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%&!"])[A-Za-z\d@#%&!"]{8,}$'

# Email Validation
if re.match(email_pattern, email):
    print("Email ID is Valid")
else:
    print("Email ID is Invalid")

# Password Validation
if re.match(password_pattern, password):
    print("Strong Password is Valid")
else:
    print("Strong Password is Invalid")

*** Enter Email ID: tharunsai@gmail.com
Enter Password: 123456789
Email ID is Valid
Strong Password is Invalid
```



```
import re
```

```
# Given passage
```

```
text = """
```

```
Artificial Intelligence (AI) is transforming industries across the world.
```

```
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,  
and in education to provide personalized learning experiences.
```

```
Many companies invest heavily in AI research because AI improves efficiency  
and enables intelligent decision-making.
```

```
As AI continues to evolve, professionals with AI skills are in high demand.  
"""
```

```
# Search for the first occurrence of "AI"
```

```
match = re.search(r"\bAI\b", text)
```

```
if match:
```

```
    print("1. First occurrence of 'AI' found.")
```

```
    # Starting position
```

```
    print("2. Starting position:", match.start())
```

```
    # Ending position
```

```
    print("3. Ending position:", match.end())
```

```
    # Count total occurrences
```

```
    count = len(re.findall(r"\bAI\b", text))
```

```
    print("4. Total number of occurrences:", count)
```

```
else:
```

```
    print("5. The word 'AI' was not found.")
```

```
... 1. First occurrence of 'AI' found.  
2. Starting position: 26  
3. Ending position: 28  
4. Total number of occurrences: 6
```

```
import re
```

```
# Compile the email pattern only once
```

```
email_pattern = re.compile(
```

```
    r'^[A-Za-z0-9][A-Za-z0-9._%+]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
```

```
)
```

```
# Number of email IDs
```

```
n = int(input("Enter the number of email IDs: "))
```

```
# Validate each email ID
```

```
for i in range(n):
```

```
    email = input("Enter Email ID: ")
```

```
    if email_pattern.fullmatch(email):
```

```
        print(email, "-> Valid Email ID")
```

```
    else:
```

```
        print(email, "-> Invalid Email ID")
```

```
... Enter the number of email IDs: 2
```

```
Enter Email ID: tharunsai123@gmail.com
```

```
tharunsai123@gmail.com -> Invalid Email ID
```

```
Enter Email ID: 192472382.simats@saveetha.com
```

```
192472382.simats@saveetha.com -> Valid Email ID
```


Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No.	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1	3	3	4	4	4	18	86.
2					3	17	
3					3	17	
4					3	17	
5					3	17	

Faculty In-charge	Course Coordinator	Program Director
<i>[Signature]</i>		
Signature: _____	Signature: _____	Signature: _____
Date: <u>25/07</u>	Date: _____	Date: _____